

BEN HORWITZ

benhorwitz05@gmail.com
ben-horwitz.com

EDUCATION

DARTMOUTH COLLEGE

Candidate for B.E. in Mechanical Engineering (*Dual-Degree Engineering Program*)

Hanover, NH
Expected May 2028

MIDDLEBURY COLLEGE

Candidate for B.A. in Mathematics
GPA: 3.88

Middlebury, VT
Expected May 2027

WORK EXPERIENCE

DESIGN AND MANUFACTURING LAB, GEORGIA TECH

Design for Additive Manufacturing Engineering Intern

Atlanta, GA
Summer 2026

- Led development of a novel SLA printer that prints high-viscosity silicones.
- Held planar tolerances of 0.001 in. across an 18 in. build plate.
- Designed custom hardware and built the PLC control software that drives it.
- Developed a custom slicer that outputs to an 8K LCD screen.

CENTER FOR ADDITIVE MANUFACTURING, SLU

Additive Manufacturing Engineering Intern

St. Louis, MO
Summer 2025

- Produced and troubleshot parts across FDM, SLS, SLA, and PolyJet platforms.
- Designed a custom reaction chamber for an organic chemistry lab.
- Led additive manufacturing outreach events at a technical college and a science center.

MECHANICAL ADVANTAGE LLC

Mechanical Engineering Intern

Middlebury, VT
Summer 2024

- Accelerated workflow by manufacturing 10 quick-release Harperizer lids.
- Programmed an automated bar-pull that removed a manual machine-tending step.
- Collaborated with three interns and 10 full-time engineers on production tooling.

TISSUE ENGINEERING SCAFFOLD FABRICATION LABORATORY, SAINT LOUIS UNIVERSITY

Research Intern

St. Louis, MO
Summer 2022

- Modified a 3D printer for synthetic matrix fabrication alongside graduate researchers.
- Authored a board-reviewed research paper and was awarded "Best Presentation."

MIDDLEBURY COLLEGE MAKERSPACE

Teaching Assistant

Middlebury, VT
Fall 2024 – Spring 2025

- Trained and supervised over 40 community members on shop equipment.
- Programmed and manufactured a golf putter on a 4-axis mill.
- Designed an escapement mechanism used as a classroom demonstration.

CAMPUS/ COMMUNITY ACTIVITIES

MIDDLEBURY ENGINEERING SOCIETY

President

Middlebury, VT
January 2024 – Present

- Spearheaded the re-branding and reorganization of the club.
- Developed and led 3D printing and laser cutting seminars for 15+ members.
- Built a lasting partnership between the club and the makerspace.

HIGHLIGHTED PROJECT

HIGH-VISCOSITY VAT PHOTOPOLYMERIZATION PRINTER

- Developed a custom SLA platform enabling fabrication of high-viscosity silicone actuators for soft robotics.
- Integrated mechanical, motion-control, and software systems into a functional printing platform.

TECHNICAL SKILLS

- **CNC machinery** (HAAS 3- and 4-axis mills, manual lathe)
- **Additive manufacturing** (SLA, SLS, FDM, RFAM)
- **CAD, CAM, FEA** (SolidWorks, Autodesk Fusion 360, Onshape)
- **Programming** (MATLAB, Python, working knowledge of R)
- **Controls** (Beckhoff PLC, Indradrive)